Ayush Amarnath Bhakat

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Education

- **Manipal Institute of Technology [MIT Manipal]** — Udupi, India
  Bachelor of Technology - Computer Science(AI & ML); CGPA: 6.73 — *2022 - 2026*
  Courses: *Operating Systems, Data Structures, Algorithms, Database Management, Object Oriented Programming, Digital Systems and Computer Organization, Artificial Intelligence, Machine Learning, Computer Vision, Artificial Neural Networks, Reinforcement Learning, Natural Language Processing, Quantum Mechanics*
- **Indian School Al Ghubra** — Muscat, Oman
  Class XII, 94% — *2022*
- **Indian School Al Ghubra** — Muscat, Oman
  Class X, 91% — *2020*

Skills Summary

- **Languages**: C, Java, Python, Swift, TypeScript, SQL
- **Tools**: NumPy, Pandas, Matplotlib, OpenCV, PySpark, PyTorch, TensorFlow, LangChain, FastMCP, FastAPI, Pydantic AI, Vercel AI SDK, Hono, Bun
- **Web Development**: HTML-5, CSS-3, JavaScript, Bootstrap, Tailwind CSS, JQuery, Django, React, Next.js, Node
- **Databases**: PostgreSQL, Firebase Firestore, Qdrant, Redis, SQLAlchemy
- **Cloud & Infra**: AWS, AWS Bedrock, GCP, Firebase, Runpod, Modal, Docker
- **Operating System**: Windows, Linux, macOS

Products & Ventures

- **LisnAI — AI-Powered Life Diary iOS App (Swift, TypeScript, Bun, Hono, Firebase, Qdrant)**: Built a commercial B2C iOS application that continuously records, transcribes, and stores conversations as a searchable life diary with actionable intent extraction and task automation. Architected a 13-agent backend pipeline (Entity, Router, Action, Memory, Chat, Query, Insight agents) using Vercel AI SDK with multi-provider LLM support (OpenAI, Anthropic, Gemini). Backend built on Bun/Hono with Firebase Firestore, Qdrant vector search, Inngest background jobs, and end-to-end encryption. iOS frontend uses SwiftUI with on-device transcription (iOS 26 SpeechAnalyzer) for a privacy-first, local-first architecture with optional cloud sync. Integrated Langfuse for LLM observability and cost tracking.
- **Aelvyn Systems — AI-Powered SDLC Automation Platform (Python, FastAPI, Next.js, PostgreSQL, Redis)**: Built a full-stack SDLC automation platform comprising 13 specialized AI agents (code review, security scanning, test generation, deployment, cost optimization, sales intelligence) orchestrated across 7 FastAPI microservices (Nexus, Forge, Sentinel, Crucible, Conduit, Herald, Ledger). Event-driven architecture using Redis Streams with 24 event types automating the PR→Review→Test→Deploy pipeline. Includes MCP server for IDE integration, CLI copilot, and Next.js admin dashboard. Three-tier LLM routing (Opus→Sonnet→GPT-4o-mini) with pydantic-ai agents, LangGraph workflows, and multi-tenant organization isolation.

Projects

- **OptiNet-GRPO: DRL-based Resource Allocation in Elastic Optical Networks (Python, PyTorch, Modal, HuggingFace)**: Developed a novel GRPO-based reinforcement learning system for routing and spectrum allocation (RSA) in C+L band elastic optical networks. Built a custom Gymnasium environment simulating NSFNET (14-node) topology with 320 spectrum slots and distance-dependent modulation selection (16QAM/8QAM/QPSK/BPSK). Designed a GQA Transformer policy (∼500K parameters) trained via Group Relative Policy Optimization with Xi fragmentation metric as reward. Implemented an LLM reward model by fine-tuning Qwen 3.5-2B on 50K+ synthetic (state, blocking_rate) pairs using SFT and GRPO via TRL. Distributed training on Modal A100 GPUs with Weights & Biases experiment tracking.
- **Mental Health Conversational Agent with Sentiment Analysis and MCP Tool Calling (LLMs, PyTorch, FastMCP)**: Developed a Multi-agent framework with MCP tool calls, Agent orchestration and chain of thought for medical analysis of user symptoms, documents like X-rays, MRI-scans and prescriptions. MCP servers were created locally using vision transformers and web-search libraries. TTS (text to speech) and STT (speech to text) modules were wrapped around the main system. Used Streamlit to run the system with a UI interface.
- **Prostate Cancer Detection (PyTorch, Matplotlib, 3D Slicer)**: Developed a Reinforcement Learning aided 3D CNN-UNet architecture to segment cancerous regions in a 3D MRI scan visualized in 3D slicer. Multiple UNet architecture was explored and created through the RL Algorithm REINFORCE, delivering an accuracy of 72% on 42 real MRI scans acquired from KMC (Kasturba Medical College).
- **Chemical Reactor Control with Reinforcement Learning (PyTorch, SkLearn, Matplotlib)**: Developed an accurate control system, monitoring chemical reactor temperatures using 2 actuators namely Coolant Flow and Heater Current. Leveraging Deep Deterministic Policy Gradient (DDPG) Algorithm and other novel improvements pertaining to control theory, NMPC and PIDs. Achieving near perfect temperature tracking trajectories through efficient roll outs on a real chemical Reactor available at MIT. This project showcased expertise in complex RL implementation, novel idea integration, brainstorming and delivering actionable results along with hands on experience of building RL models for a real (unsimulated) Chemical Reactor.

- **Replicated Claude Code CLI (Python, Rich, LangGraph, FastMCP, FastAPI)**: Built a command line interface called DUNE CLI inspired by Claude Code which automates code generation, file creation, debugging and self-improving code. Using LLMs as the main orchestrator agent, this CLI has 7 tools which the agent can reason and use for the task specified by the user within an interactive chatbot like session. Core features that make Claude Code stand out have also been replicated: Code-base Indexing, Retrieval Augmented Generation (RAG), Chat Session memory, Auto looping and recursive tool call with user permission options for bash command. This project showcased expertise in rapid adoption of AI and replication of emerging technology which includes production ready projects.
- **Constructing Small Reasoning Model using Knowledge Distillation and GRPO (Python, SkyRL, PyTorch, FastAPI)**: Built a Small Language Model with 3B parameters (inspired by the Qwen 32b MOE architecture) using PyTorch to create Grouped Query Attention (GQA) with efficient expert routing to enable Mixture of Experts (MOE) and trained using Runpod A100 GPUs. Leveraging Knowledge Distillation (teacher student learning) Knowledge was transferred from Qwen 32b teacher model into the self-made SLM 3B MOE model after pretraining. Reasoning was further enabled using the RL technique called Group Relative Policy Optimization (GRPO) for MOE models which induces $< think >$ tags. This project showcased the feat of engineering needed to create SLMs from scratch, the expertise in multi-disciplinary skills with Deep Learning and Agentic AI along with GPU optimization skills on cloud-based platforms.

Experience and Extracurricular

- **AI Intern — Phronetic AI, Neuromind Technologies Pvt. Ltd. (Onsite, Bengaluru)**: Jan 2026 - June 2026
  - Phronetic AI is India's first full-stack Agentic AI platform, backed by Infibeam Avenues Limited, enabling low-code design, configuration, and deployment of AI agent teams for enterprises.
  - Designed and developed a Context-Aware AI Copilot with progressive tool disclosure for automated agent configuration. Built 3 architectural versions: a foundation with 6 core tools and SSE streaming middleware (JSON Patch state sync, reasoning extraction, tool-name humanization), an expanded version with 15+ tools featuring automated LLM-driven tool creation with AST validation, API import parsing, and MCP-based third-party SaaS integration, and a production-hardened version introducing a Skills architecture for dynamic tool visibility via deterministic guard functions, modular prompt assembly from 4 independent files, and multi-pattern leaked output suppression. Core stack: Pydantic AI, LiteLLM, AWS Lambda, MCP.
  - Led a cross-functional team as Project Lead to acquire the Gujarat Police Department as a client. Built an end-to-end AI pipeline for FIR filing automation that intelligently predicts applicable legal sections (IPC/BNS) to prefill FIRs using NLP-based classification. Delivered the product pitch and technical demonstration to the client, resulting in successful acquisition.

- **AI Intern — C-HIT (Remote)**: July 2025 - Sept 2025
  - C-HIT offers healthcare IT solutions leveraging cybersecurity and AI/ML techniques to modernize federal health systems.
  - Built EVIQ, a production-ready EDI X12N 278 validation engine (FastAPI, 2200+ line validator) implementing WEDI SNIP Level 1–7 compliance testing with configurable segment terminators, automatic encoding conversion, and HIPAA-compliant audit logging. Developed a comprehensive ISA/IEA envelope validator and dual-schema comparison for custom trading partner specifications.
  - Developed a multi-agent AI system (LangGraph, Groq/OpenAI) with three specialized agents: an Enhanced Validator Agent with sub-agents for syntax intelligence, business rules (TR3), and ICD-10/CPT/HCPCS code set validation; a Dynamic Mapper Agent for automated ESL schema generation and field mapping; and a Document Generation Agent for compliance reporting and executive summaries.
  - Built a RAG pipeline using ChromaDB vector database with EDI-specific chunking strategies and Neo4j knowledge graphs for healthcare terminology relationships, enabling semantic search across EDI standards, error patterns, and compliance rules for context-aware validation.
  - Implemented automated user story and test case generation from business requirements, and automated EDI file generation using auto-diff comparison against X12 standards for verification and validation workflows.
  - Designed the prior authorization automation system aligned with the CMS WISeR Model, integrating FHIR R4 interoperability for medical necessity assessment, claims prediction, and denial risk analysis.

- **AI Intern — Oman Data Park (Onsite)**: Dec 2024 - Jan 2025
  - ODP is a cloud service provider offering datacenters, cybersecurity and AI-enabled cloud solutions for enterprises and government agencies.
  - Developed a Multi Agent system for the company data, leveraging MCPs tools like web-search, OCR tool, custom made MCPs for Sub-Agents along with chain-of-thought reasoning and multi-agent orchestration to effectively process company data and query complex task. A RAG based system was used and the prototype was run with Streamlit UI.
  - The prototype was used for internal querying and testing as the company explored various ideas around improving their AI domain (No access to sensitive data was granted).

- **Ather Hackathon**: April 2025
  - Ranked top 30 among all over India participants for Ather's ByteBattles2.0 All India Firmware Challenge. Qualified for the final round demonstrating competence and multi-disciplinary skill set in electronics and Machine Learning.

Achievements and Certifications

- Ranked Top 30 among 700+ all over India participants for Ather's ByteBattles2.0 All India Firmware Challenge, 2025.
- Completed Quantum Computing on campus drive gaining valuable insights into Quantum computing for Neural Networks and earned a participation certificate.